SCI-VAL

Sample and Detector Validity, Flags, and Map Eligibility

Producer Facts and Use-Specific Eligibility: Not Final Map Validity

Science-Team Rationale

Scientific contract version	v0.1
Rationale revision	r0.3
Date	2026-08-24
Input	Exact immutable producer facts/supports plus one exact owner-bound registered profile
Output	Cause-preserving, use-qualified request/applicability/eligibility/ realization record
Governing rule	Producers own facts; named-use owners own policy; VAL Core evaluates; the registry binds but does not author policy
Source	Owner-approved r0.3 Core directive plus exact continuing source and profile registries
Status	Content-bound freeze candidate; scientific-owner approval pending; implementation conformity and validation not assessed

Central claim. VAL is an evaluator and registry mechanism, not a policy author. Registration records who owns a policy, exactly which source defines it, and how it may be replayed; it never grants scientific authority to VAL.

Executive abstract. A finite value is not necessarily an independent measurement. A retained value is not necessarily admissible to a fit. A value admitted upstream is not necessarily projected into a map. SCI-VAL keeps these questions separate by combining exact producer-owned facts with one exact policy owned by the scientific use. Its four axes distinguish whether a question was requested, whether it applies, what its scientific disposition is, and whether an authoritative decision artifact was realized. Causes remain visible; silence is not evidence of absence; and an unrelated permission cannot rescue an applicable exclusion.

V0.1 has one mandatory canonical registered profile: `SCI-VAL:independent_exposure@1`. Its direct-origin rule is a contract-level scientific invariant: a synthesized or replaced representative occurrence is not an original independent exposure, and no exception may redefine it. The continuing registry also binds five exact PTC-owned profiles for basis fit, loading fit, frozen operator application, ordinary output retention, and existing tracked-kernel propagation. Generic coefficient/QC and empirical/simulation identities are explicitly unsupported; MAP remains unbound. The companion Engineering Conformance Specification is the complete VAL Core formal authority; the continuing registries bind owner policies and sources without rewriting that Core.

1 The finite, retained, replaced occurrence

Start with one exact detector occurrence. Its stored payload is finite and is retained. RTC states authoritatively that the exact representative source was replaced and supplies the replacement cause and source link. The following questions are not aliases.

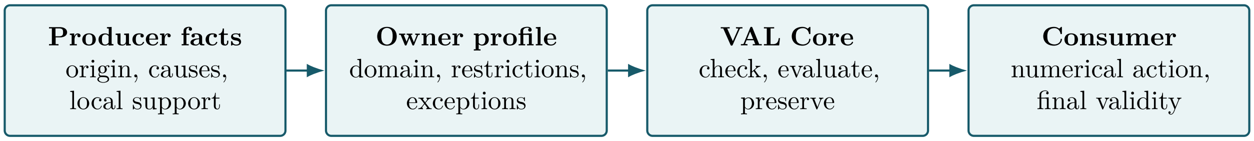
Question	Profile/authority status	Scientific outcome
Independent astronomical exposure?	Mandatory canonical profile plus RTC origin fact	**Ineligible.** Direct replacement is non-exceptionable; the cause remains visible.
Receive a frozen operator?	Exact `SCI-PTC:operator_application@1` profile; required group-time facts are not supplied by this worked case	**Decision unavailable here.** Direct replacement is not itself an application veto; complete operator facts would decide this separate use.
Remain in an ordinary PTC output?	Exact `SCI-PTC:output_retention@1` profile plus RTC replacement fact	**Ineligible.** Direct replacement is excluded from ordinary PTC science-signal support even if a numerical value was calculated.
Display for diagnosis?	Purely informational display; no admission or decision role requested	No eligibility proposition. The value may remain visible with replacement provenance and no stronger claim.
Enter PTC basis fitting?	Exact `SCI-PTC:basis_fit_admission@1` profile plus RTC replacement fact	**Ineligible.** The occurrence teaches neither ordinary centering nor the PCA basis.
Enter PTC loading fitting?	Exact `SCI-PTC:loading_fit_admission@1` profile plus RTC replacement fact	**Ineligible.** Loading permission remains distinct from basis permission.
Enter generic coefficient/QC estimation?	Broad profile identity explicitly unsupported	No present proposition is available for VAL to evaluate; a future diagnostic with decision authority needs its own exact profile.
Pass MAP upstream admission?	MAP boundary known; exact profile absent	Decision unavailable. Even eligibility would not prove projection, contribution, support, or final validity.

Why one flag fails. This occurrence is finite and upstream retained, but directly replaced and therefore ineligible for independent exposure, ordinary PTC basis/loading influence, and ordinary PTC output retention. Exact frozen-operator application remains a separate question; informational display asserts no eligibility proposition; MAP remains unbound. A single good/bad bit loses the scientific question, owner, and reason.

The independent-exposure result is not a hidden VAL policy. The exact scientific owner is recorded in the immutable registry binding; the `SCI-VAL` namespace identifies the registry package, not the policy owner. VAL checks and evaluates that immutable binding. A policy that tries to override the non-exceptionable origin rule is invalid. Hypothetical complete future profiles remain unavailable until their exact owner, source, domain, restrictions, and compatibility are registered; a name alone conveys no permission. "Would be eligible" is therefore a conditional statement about a complete future binding, never a present decision. The invalid-origin override yields decision unavailable with a policy-invalid reason, never eligible.

2 Four authorities must stay separate

The scientific chain has four distinct authorities. A producer describes what happened and what its own operation supports. The owner of the named use says which of those facts matter for that exact question. VAL checks identities and executes the supplied rule. The consumer performs its numerical action and owns the final product claim.



The Profile Registry sits alongside this chain. It binds an immutable profile identity to the actual scientific owner, source version or digest, domain, restrictions, exception permissions, and compatibility rules. It does not copy the policy into VAL ownership. A reserved name with no exact binding is not a usable policy.

Authority	Owns	Cannot infer
Producer	Facts, causes, explicit negative assertions, producer-local composites and influence	A downstream use policy
Named-use owner	Applicability, restrictions, exceptions, aggregation, thresholds, failure scope	Facts owned by another producer
Profile Registry	Immutable owner/source/domain/compatibility binding	A missing predicate or owner
VAL Core	Shared logic, provenance, deterministic evaluation	A PTC or MAP policy, threshold, or universal flag
Consumer	Estimator action and final product validity	Promotion of an invalid parent because a descendant is finite

This split is why PTC owns basis-fit, loading-fit, application, output, and coefficient/QC supports separately, while MAP owns upstream admission, projection, numerical contribution, support, response/covariance, coaddition, and final validity separately. VAL can evaluate a registered profile for one of those questions; it cannot collapse the questions or author the answer.

3 Four axes, open-world facts, and no-rescue logic

Every decision carries four independent answers.

Axis	Values	Question answered
Request	requested / not requested	Was an authoritative evaluation asked for?
Applicability	applicable / inapplicable / unknown	Is this profile defined for this object and domain?
Eligibility	eligible / ineligible / decision unavailable	What can the known facts establish for this exact use?
Realization	realized / incomplete / failed / not produced	Does an authoritative decision artifact exist?

If no request was made, no eligibility proposition is invented. If the profile is known not to apply, there is likewise no eligibility proposition. If the scientific question is well defined but a required fact is unresolved, the correct result is decision unavailable. A failed output artifact is a failure of realization, not a new scientific disposition.

Facts are open-world. An authoritative producer can say true or explicitly false; information can also be unknown or conflicting. Silence about a cause is unknown. It becomes authoritative absence only when the owning producer declares the relevant cause family complete and explicitly asserts absence.

A conflict in identity, parentage, profile authority, applicability, or another structural-gate fact prevents the scientific question from being established: applicability is unknown and the decision is unavailable. Once the domain is established, a non-gating conflict is preserved as unresolved evidence rather than promoted to a structural failure.

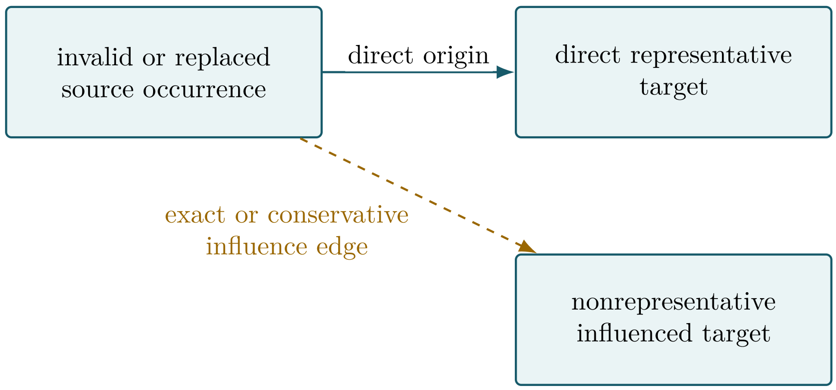
Once the structural question is established, applicable restrictions compose conservatively:

$$\begin{gathered}
\text{one known exclusion} \quad + \quad \text{any unrelated permission, unknown, or conflict} \quad \Longrightarrow \quad \text{ineligible,} \\
\text{no exclusion} \quad + \quad \text{at least one required unknown or conflict} \quad \Longrightarrow \quad \text{decision unavailable,} \\
\text{all required permissions known} \quad \Longrightarrow \quad \text{eligible.}
\end{gathered}$$

A permitted exception can neutralize only the exact exceptionable restriction named by the same owner-bound profile. It never erases the underlying cause, and it cannot touch the independent-exposure origin invariant. Unknown or conflicting exception applicability cannot neutralize the restriction; the underlying exclusion remains and the conflict is preserved. A permission is a use-specific disposition, not a new cause.

4 Direct origin, inherited influence, and immutable generations

Direct origin and inherited influence answer different causal questions. A representative occurrence synthesized or replaced upstream is directly not an original independent exposure. A different occurrence may only be influenced by that source through an operator. That edge can be exact and confirmed, or conservative and merely possible. A conservative envelope may deliberately overstate influence; it must never be relabeled exact.



The direct representative link activates the independent-exposure invariant. The nonrepresentative edge has no universal downstream disposition. Its use owner may permit it, exclude it, request review, or state scientific unavailability. Review is action metadata, not a fourth eligibility state, and no choice changes the producer-owned causal record.

Decisions are snapshots. Later information creates a later generation:

$$\text{facts}_k \longrightarrow \text{VAL decision}_k \longrightarrow \text{consumer action}_k \longrightarrow \text{facts}_{k+1}.$$

The later generation cannot rewrite what the earlier consumer knew or did. This is especially important for staged PTC decisions. A later output reject does not rewrite basis-fit membership; a coefficient/QC decision does not rewrite a loading fit; only the PTC owner can decide that a changed fit support requires refitting.

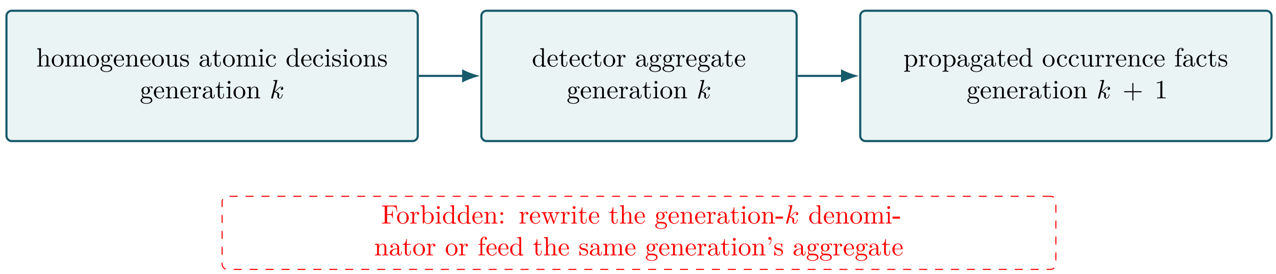
Source and profile changes follow the same rule. A new source digest, exception resolution, profile version, or compatibility binding creates a new replay identity. “Use the current policy” is not reproducible scientific provenance.

5 Aggregation is a new scientific proposition

An aggregate such as a detector-level rejected fraction is not a bare number. It has its own registered profile identity, scientific owner, version, applicability domain, operator, threshold, and propagation authority. That aggregate profile also binds the exact atomic source-profile identity from which its inputs were produced. It cannot reuse the atomic profile identity or masquerade as another atomic occurrence. Its owner must additionally state the exact occurrence population and time support, counts on all four axes, denominator, treatment of missing decisions, boundary polarity, uncertainty, and whether the result is advisory or binding.

For example, the atomic profile might be `SCI-VAL:independent_exposure@1`, while a future aggregate could be `SCI-PTC:detector_independent_exposure_fraction@1`. The latter name is hypothetical and unavailable here; only a complete immutable record from its actual owner could make that aggregate proposition evaluable.

Base v0.1 adds a strict compatibility rule. Every atomic decision must share the same exact profile identity and version, lifecycle stage, object type, and applicability domain, and the aggregate profile must name that exact compatible atomic source profile. "Eighty percent eligible" is meaningless if some inputs were judged for diagnostic display and others for independent exposure. A heterogeneous aggregate is unavailable unless a future owner-approved transformation maps all inputs to one common proposition; VAL supplies no such transformation.



Reverse propagation therefore creates new producer facts and new decisions at generation $k+1$. It cannot alter the generation-k decisions that formed its own denominator. Any iterative detector classifier needs separately approved bounds and a stop rule.

An empty or unknown denominator is never a valid zero fraction. Permuting the same homogeneous multiset cannot change the result. Partitioned and unpartitioned evaluation are equivalent only when the aggregation owner has declared a sufficient summary and associative combine rule for the exact missing, threshold, boundary, and uncertainty semantics.

6 Profiles, source bindings, response, and uncertainty

Package-qualified names stop broad vocabulary from merging different scientific questions. V0.1 reserves distinct PTC names for basis-fit admission, loading-fit admission, frozen-operator application, output retention, coefficient/QC population, response companion, and empirical/simulation population. MAP uses `SCI-MAP:map_upstream_admission`, not the ambiguous former phrase `analysis_or_gridding_contribution`. A reserved name still supplies no policy.

The table below is the **r0.3 registry snapshot**. `SOURCE_BINDING_REGISTER.md` is the continuing authority, so a later adjacent binding update does not by itself rewrite VAL Core.

Source	Exact binding used here	Availability consequence
RTC	v0.1/r0.9 frozen authority plus sanitized boundary	Representative origin and influence meanings are bound; changed origin semantics require a new compatibility review.
CAL	v0.1/r0.3 architecture-frozen rationale; scientific authority open	Boundary meanings are conditional; missing numerical authority remains unavailable.
PTC	v0.1/r0.4 frozen authority as recorded in the program index plus sanitized boundary	Stage meanings are bound; exact PTC policy profiles are not supplied by VAL.
MAP	v0.1/r0.3 house rationale; scientific authority open; accepted F010/ADR boundary	Boundary separation is bound; exact MAP upstream-admission profile remains unavailable.

Response and uncertainty availability are typed facts, not numeric zero or identity. The use owner assigns exactly one role:

Role	Scientific consequence
`structural_gate`	Without authoritative satisfaction, applicability is unknown and the decision is unavailable.
`required_permission`	Availability is required; known failure excludes, while unknown or conflict is unresolved.
`decisive_exclusion`	Authoritative failure excludes; unknown or conflict is unresolved.
`advisory`	The state is preserved but cannot determine eligibility.

VAL executes that role but does not choose it. Invalid payloads are excluded before arithmetic. If an admitted required payload is then found non-finite, the declared operation fails or becomes unavailable at the owner-defined scope. Finiteness never repairs missing calibration, response, uncertainty, parent, or policy authority.

Availability is not rejection. An unavailable profile, source, response, or required fact yields an explicit inability to decide. It is not silently interpreted as clean, zero, identity, or eligible.

7 What would test the contract, and what remains deferred

The engineering specification states 49 normative requirements and 24 falsifiable predictions. They define future checks; they do not report that an implementation has passed.

Evidence layer	Question it could answer	Not established by this draft
Logical fixtures	Does the evaluator preserve axes, causes, registry bindings, truth-table outcomes, and deterministic replay?	Present implementation conformity
Representation checks	Does a chosen schema preserve every required distinction and immutable identity?	Scientific correctness of thresholds
End-to-end scientific tests	Do known injections, source changes, aggregate partitions, and propagation generations behave as predicted?	Observational performance without data
Observational validation	Do real reductions meet preregistered scientific criteria?	Production readiness by itself
Operations review	Are evidence, failure handling, monitoring, and ownership adequate for use?	Authority to change the scientific contract

No general SCI-VAL scientific question remains open from the six first-pass derivation items. Science now fixes the partial eligibility proposition, conflict precedence, review as action metadata, the four response/uncertainty roles, exception lineage, and the conservative aggregation rule. Exact serialization of an uninstantiated eligibility slot and review metadata is engineering-deferred. Sufficient summaries, associative combine rules, and partition equivalence remain profile-local declarations, not VAL defaults.

The main falsifiers are concrete. A direct replacement admitted as independent exposure, a reserved name yielding eligibility, a mixed-profile aggregate being treated as a valid fraction, propagation rewriting its own denominator, a changed source digest replaying as the old policy, or MAP upstream admission being promoted to actual contribution would each violate this contract.

Present status. This r0.3 revision resolves the general architecture, authority, naming, aggregate-identity, conflict, role, feedback, and source-binding questions assigned to this surgical round. It does not freeze SCI-VAL, assess an implementation, validate observational behavior, or authorize production use.

Reading rule. The companion Engineering Conformance Specification is the complete formal authority. This rationale explains that authority in science-team language and cannot waive a requirement or convert an unavailable state into a pass.